Classification and gender recognition from veiled-faces

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Abstract: This study aims to investigate to what extent a computer system can identify veiled-human and recognise gender using eyes and the uncovered part of the face. For the purpose of this study, we have created a new veiled persons image (VPI) database shot using a mobile phone camera, imaging 100 different veiled-persons over two sessions. After preprocessing and segmentation we used a fused method for feature extraction. The fusion occurs between geometrical (edge ratio) and textural (probability density function of the colour moments) features. The experimental results using different classifiers were ranging from 88.63% to 97.22% for person identification accuracy before feature selection and up to 97.55% after feature selection. The proposed method achieved up to 99.41% success rate for gender classification.

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1 Introduction

In the past few years, automatic face recognition has become one of the most significant biometrics authentication technique (Li and Jain, 2011; Chihaoui et al., 2016). Despite the importance of the human face for social interaction and person identification, some people in the population insist on veiling their faces for certain cultural and other

purposes. In real life, if there are computer applications that can identify a veiled-face and recognise the gender, it will be a greatly valuable and beneficial for different security communities. Such areas of interests include but not limited to, facilitating the identification process of Muslim-women who get dressed Niqab (a cloth that covers the face in exception eyes and the surroundings. It is a part of sartorial hijab and worn by some Muslim women in public areas especially in the Middle Eastern countries), to identify them without asking them to reveal the whole face, in criminal investigation, to identify veiled-terrorists, in situations where women are impersonated by men wearing Niqab, and in general for occluded face recognition applications.

Biometric recognition based on the whole face or periocular region is a widely researched area with many different techniques proposed in the last few decades (Krotewicz et al., 2015; Moreno et al., 2016; Teo et al., 2007). Nevertheless, identifying a human depending on a part of the face (in case of veiled-person), and using a camera is a challenging task, and has, to the best of our knowledge, never been investigated in the literature. In particular, the periocular and ocular biometrics are concerned with finding identification features around the eye or inside the eye respectively. In contrast, our work here takes both the eyes, and areas around the eyes such as eyebrows and the upper part of the nose; hence this work falls under neither periocular nor ocular biometrics. However, our approach is related with partial face biometrics (Teo et al., 2007) that relies on a part of the face. Nevertheless, the experiments in Teo et al. (2007) were conducted on a different face database (namely the Essex database), and they cropped the face image taking the rectangle of the eyes area, this approach is different from the faces are cropped by the proposed method here is instead taken from a real veiled persons, with different subjects. The novelty of this work is about recording a new database for veiled persons, and investigating image processing methods to see how much to the extent that we can identify those persons. That is, the main purpose our work is to investigate to what extent an automatic computer system can identify a veiled-person just from the revealed part of the face alone; that is the eyes and the surroundings. For the purpose of this study, a new veiled-persons image (VPI) database has been created, consisting of veiled-face images of 100 different adult individuals (both male and female)[1]. This database is used in our investigation, particularly for training and testing the methodology.

The rest of this paper is organised as follows. Section 2 presents some of the previous work related to face recognition and gender classification through face images. Section 3 describes the database used for this investigation, presents the methods used for preprocessing, segmentation, and features extraction. Section 4 presents and discusses the results of the proposed methods and compares them with some methods in literature. Finally, in Section 5 conclusions and future work related to the study are provided.

2 Brief literature review

Face Recognition has several advantages over other biometric modalities: besides being natural and nonintrusive it can be captured at a distance and in a convent manner. According to Hietmeyer (2000) facial features scored the highest capability in a machine readable travel document (MRTD) among the six biometric attributes that were considered. Different methods have been proposed in the literature for the face

recognition (Chihaoui et al., 2016). Local binary pattern (LBP) descriptors are used widely in the literature because it is a fine scale descriptor that captures small texture details (Boutellaa et al., 2015). These include the work of Sharif et al. (2011) method based on the theory of Sobel LBPs and Laplacian filters edge detectors to represent face images. Thereafter, they used discrete cosine transform (DCT) to represent these face images in low dimension space. Their method achieved about 96% accuracy. DCT is used in the recognition system which presented in Dahmoun et al. (2016). Their technique is based on the global 2D-DCT, local LBPs, with recognition rates 95.5%, and 100% obtained for ORL, and Yale databases respectively. In Sharma et al. (2013), the face image was transformed by Gabor wavelet, and then divided into nine subsections. Then LBP was applied on each subsection to form the feature vector. The achieved recognition efficiency was 98.57% on ORL, FERET and LFW face recognition databases[2].

Various modifications involving the LBP have also been considered in the recent past. For example, Hu et al. (2016) in their work consider LBP feature extraction-based edge face detection and combined with full search (FS), and three-step search (TSS) image search method; to search for a face texture blocks. The modified LBP improved the recognition rate up to 92.3%. Another class of approaches are based on multiscale and localised transformations. Wavelet transformation is one of the most popular of the time-frequency-transformations; it is widely utilised in the field of biometrics area in general and face recognition in particular. The proposed method in work of Sagar et al. (2015) involves face recognition by using convolution, discrete wavelet transform (DWT) and vector compression. Their method achieved 93.33% accuracy on the standard databases such as ORL, JAFFE, Yale and Indian male and Indian female. Li et al. (2015) joined pixel-level and feature-level fusion at the top-level's wavelet sub-bands for face recognition. Their technique makes full use of all top-level's wavelet sub-bands; because the high frequency sub-bands (especially the top-level) contain some essential information that are identical to the edge and contour of face. The highest accuracy was achieved is 99% on FERET, ORL and AR face databases. Hamidi et al. (2015) based on local dual-tree complex wavelet transform (DT-CWT) to extract the features of the face image. Their method succeeded in recognising 95.5% of FERET[3] and 96.66% of Yale databases.

Related to our present work, gender classification from face images have been investigated in literature previously. Akbar et al. (2015) introduced hybrid features for gender classification by using transformation methods such as DWT and discrete sine transform (DST). To extract numerical features from face images LBP and local phase quantisation are used to achieve 92.1% success on Stanford University medical students (SUMS) database[4]. Another related system presented in the work of Ullah et al. (2014) used LBP and spatial features. They employed two texture descriptors, namely the classical LBP, Weber's law descriptor (WLD), and investigated their spatially enhanced versions (SLBP and SWLD) to distinguish male from female. Their system achieved 99.33% accuracy on FERET database. We refer to Santarcangelo et al. (2015) for a recent review on gender recognition.

From the above methods, we note that there are encouraging results with high accuracies for human and gender recognition from the human face image. However, all of these studies based on the image of the whole face whether they try identify humans or to classify genders. In our present study, we propose a fused features method to identify persons from only part of a visible face. Further, the person can be classified

into male or female from the same revealed part of the face. In addition, the total number of features in the proposed method – 115 features – is less than those typically used in the previous studies, thereby making our approach attractive with less complexity.

3 Data and methods

3.1 VPI database

We first describe our test dataset which is used in testing our method as well benchmarking other related recognition and gender classification methods. A mobile phone camera (41 mega pixel, $3{,}264 \times 2{,}448$) was used to capture the veiled-face images of 100 persons (male and female) from different age groups in the range 13 to 50 years. The images were taken in two different sessions with a different gender images collected in each session. The total number of images is 1,200, and all the images of the veiled faces were of frontal pose. We scaled down all the images in VPI to be (0.25) of the actual size, this was done to reduce the computations and increase the speed of our experiments. This new VPI was designed mainly for the purpose of this study, which is to identify persons from their eyes and the appeared part of the face. However, it can also be used for other purposes such as distinguishing between male and female, which is a much harder problem.

We followed the following rule for naming the image files: The file name starts with the session number (session 1 – S1 or session – S2), then (P) followed by the person's number, then the subject gender (male – M or female – F), which is followed by the age (in years) of the person, and finally the image number within each session ranging from 1 to 6. Table 1 depicts some examples of the naming system of the image files employed in our VPI database.

Table 1 Our VPI database consist of the following details

#example	*Session*	*Person*	*Gender*	*Age*	*Image*	*File name*
1	S1	P1	F	38	1	S1-P1-F-38-1
2	S1	P10	F	18	6	S1-P10-F-18-6
3	S2	P20	M	23	3	S2-P20-M-23-3
4	S2	P99	F	37	2	S2-P99-F-37-2

3.2 Methods

A typical face biometric system consists of a sequence of several major steps, including: image acquisition, face segmentation, feature extraction, training/testing and storing the templates of the subject for identification result, see Figure 1 which shows these important steps.

We follow a similar pipeline and describe here in detail the steps related to veiled-faces classification and recognition. In the first step, we acquired images and created the VPI database. In the second step the images were resized into 0.25 of the original size in order to get more faster processing, and then were converted to

greyscale. Object segmentation as a pre-processing step plays an important role for different applications (Shi et al., 2015). Diverse approaches and algorithms have been proposed for image segmentation. However, it is still difficult to evaluate whether one algorithm produces more accurate segmentations than another, whether it is for a particular image or set of images (Hassanat et al., 2017). To assess the accuracy of a segmentation method visually compared or human evaluators are required to look at the image segmentation results[5].

Figure 1 Flow of typical face recognition systems

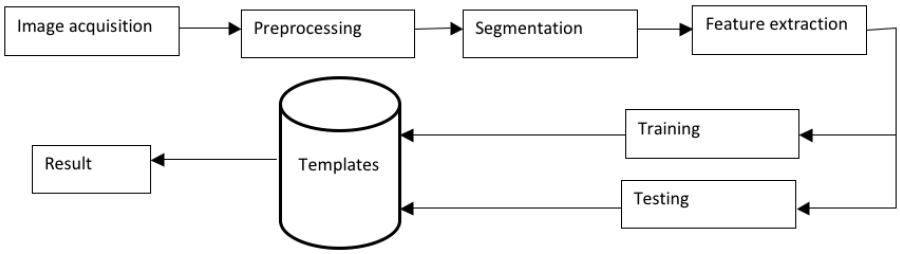


Figure 2 Segmentation results for an example image from the VPI database, (a) input image (b) Otsu thresholding method (c) global threshold (d) global RGB threshold (e) KNN (see online version for colours)

(a) (b) (c) (d) (e)

In this paper, we examined three different approaches to segment the VPI image as a preprocessing step,

1 Otsu's thresholding method (Otsu, 1979)

2 the Global thresholding based on the colour information (RGB colour model)

3 Global thresholding based on the greyscale information (greyscale model)

4 face segmentation based on colour information using ANN (Hassanat et al., 2015).

We applied all mentioned algorithms on all of the images in VPI; a sample of the segmentation results is shown in Figure 2. As can be seen from Figure 2(b) Otsu's method found to be the best segmentation for most of the images in VPI; so we had used it to segment images in this work. This assertion was backed up by subjective visualisations, that is, by examining the output of the segmentations one by one, and since there are no ground truth segmentations, we had to do it subjectively without using a quantitative method.

3.3 Features extraction

In the literature global and local features have both been demonstrated to be complementary for robust face recognition and verification (Zhou et al., 2013). In this work, we decided to restrict to local features due to the availability of a small region around the eyes in the face. We investigated two separate methods for extracting distinctive features to be used for identification.

3.3.1 Method 1 – Geometric features

Geometric features are features of objects constructed by a set of geometric elements such as edges. Edges are very important as they represent the boundary of different image regions; so they seem to be a suitable descriptor for the veiled-human eyes. To extract a geometrical feature vector from the image, three main steps are done sequentially after the preprocessing segmentation process: Sobel edge detection from the grey image, split the image into sub regions, and then compute the edge ratio for sub region. The feature vector is composed of edge ratio of all sub regions. The details are described in Algorithm 1, and Figure 3 illustrates the preprocessing steps of the feature extraction process and Sobel edge detector output.

Algorithm 1 Finds edge ratio vector of input image

Input: Veiled-face image
Output: Feature vector of edge ratio

```
Read img1 file from a folder of the VPI database
img2 = greyscale (img1)
img3 = segmentation (img2)
img4 = (using img3 as a mask and depending on the largest white area in it) crop region
    of interest (ROI) from img2
img5 = Sobel edge detection (img4)
Split img5 into 100 sub regions with dynamic size (sub region size is related to the
    image size).
Scan img5 and for each sub region
procedure Edge ratio
    If the pixel value > 32 (this means edge pixel) add 1 to number of edge pixels
    Edge ratio = number of edge pixels divided by sub region size
    Add edge ratio to feature vector
end procedure
```

Figure 3 Steps for geometrical feature vector computations from an example image in the VPI database, (a) original (b) greyscale (c) segmented (d) cropped ROI (e) Sobel edge detector (see online version for colours)

(a) (b) (c) (d) (e)

3.3.2 Method 2 – Texture features

Texture features are powerful face descriptors because they are believed to be more robust to variations of facial pose, expression, occlusion, etc. (Choi et al., 2012). In this work, we analyse the random texture of the face image by three statistical values; mean, variance and skewness. The first three moments (mean, variance, skewness) for colour value ($V(i,j)$) and block size (N) are defined by

$$\text{Mean} \qquad \mu_i = \frac{1}{N}\sum_{j}^{N} V(i,j) \tag{1}$$

$$\text{Variance} \qquad \delta_i = \frac{1}{N-1}\sum_{j}^{N}(V(i,j)-\mu_i)^2 \tag{2}$$

$$\text{Skewness} \qquad \gamma_i = \left[\frac{1}{N}\sum_{j}^{N}(V(i,j)-\mu_i)^3\right]^{\frac{1}{3}} \tag{3}$$

After cropping the ROI from the grey image, the image is divided into blokes. For each block mean, variance and skewness values are calculated to form three separated vectors, and then the probability density function (PDF) is calculated for each vector; the steps of calculating the PDFs shown in Algorithm 2. Finally the three PDF values are merged to form the final feature vector. Figure 4 represents the flowchart of the feature extraction in method 2. As seen in Figure 4, we need to get the final PDFs in order to form the final feature vector. The following steps are to be applied to calculate the PDFs:

1 After extracting the three vectors (mean, variance and skewness) for each block.

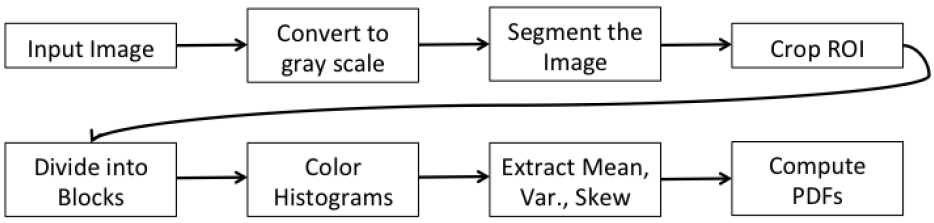
2 Create histogram (binning) for each one of these three vectors. That means, divide each vector into intervals; each interval contains a number of values that belong to that interval.

$$\text{Interval length} = \frac{\max(\text{vector}) - \min(\text{vector})}{\text{bins number}} \tag{4}$$

3 To create a PDF each value of the histogram is divided by the sum of histogram values (the number of blocks because there is one value from each block).

4 The PDFs are concatenated as one feature vector to form the final feature vector. The steps of computing PDFs are explained in Algorithm 2.

Figure 4 illustrates the steps of the feature extraction in method 2



Algorithm 2 Finds PDF values

```
Input: Moment's vectors (mean, variance, and skewness), bins number
Output: Feature vector of PDFs
Read bins number
for each moment vector do
    Read the vector values
    Create the histogram (binning) of the vector:
    Find min (vector), max (vector)
    Compute interval length (INTL) = (max(vector)-min(vector))/(bins number)
    Sort values in the vector
    Divide the vector into intervals with INTL length
    Count the values that belong to each interval
    Add the counted number to the histogram vector
end for
Create PDF vectors:
for each histogram vector do
    Read the vector values
    Calculate PDF value: divide each value by block numbers
    Add PDF value to PDF vector
end for
Concatenate PDF vectors to form the final feature vector
Return final PDFs vector
```

4 Experimental results and discussion

To evaluate the identification process for persons in VPI we conducted four major experiments.

1 identification of persons using method 1 – geometric features.

2 identification of persons using method 2 – texture features.

3 identification of persons using concatenated features from both methods 1 and 2.

4 we tested the method that scored the highest accuracy on VPI to classify individual's gender.

We worked on both Session_1 and Session_2 parts from VPI database using several classifiers. We partitioned the VPI database randomly into ten equal sized sets. Of the ten sets, a single set is retained as the test data for testing the model, and the remaining nine sets were used as training data. This process was then repeated ten times (ten folds cross validation), with each of the ten sets used exactly once as the test data. The ten results from the folds were averaged to produce the final result. We also tested different classifiers: artificial neural networks (ANN), K-nearest neighbours (KNN), random forest (RF), multiclass, and naïve Bayes classifiers.

In the first experiment, we investigated the effect of the sub region size on the identification accuracy when applying method 1 – geometrical features. Three sizes for sub regions were investigated:

a 25 sub regions: image width/5 × image height/5

b 50 sub regions: image width/5 × image height/10

c 100 sub regions: image width/10 × image height/10.

Table 2 shows the variety in accuracy of method 1 related to the sub region metrics using different classifiers. It can be seen from the table that there are two major outcomes:

1 method 1 recorded the highest success rate (95.96%) for person identification when the sub region width = image width/10 and its height = image height/10

2 using ANN classifier scored the best accuracy to identify person.

The main reasoning for this result is the way the division was made, that is when the image is divided into smaller sub regions, the extracted features increased and we get more detailed geometrical information about the image which in turn implies better accuracy.

Table 2 Accuracy of person identification in VPI database using geometric features with different classifiers (method 1 – geometrical features) with different size of sub regions

Classifier sub-region size	*ANN hidden layers = 3*	*KNN Euclidian dist. K = 1*	*KNN Manhattan dist. K = 1*	*RF batch size = 100*	*Multiclass 1-against all*	*Naïve Bayes batch size = 100*
25	93.35%	92.34%	92.14%	91.58%	88.72%	66.58%
50	93.43%	93.26%	93.35%	91.84%	91.66%	81.31%
100	**95.96%**	94.05%	95.79%	93.60%	93.93%	81.90%

The performance of method 2 was examined in the second experiment. Bins number and block size are the main parameters in this method; therefore, we examined two separate cases:

1 bins number is tuned to 5

2 bins number is tuned to 10.

Because we have 3 PDFs, one for each of mean, variance, and skewness; the final feature vector size is 15 in the first case and 30 in the second case. The block size is determined dynamically depending on the image size; because we implemented partitioning of the image into 100 blocks regardless of the image size. We used 100 blocks, because it performed the best in the previous experiment, as can be seen in Table 2 last row. Accuracy rate of person identification in two mentioned cases is shown in Figure 5.

As can be seen from Figure 5, the top accuracy of 84.59% was obtained with the number of bins = 10 on RF classifier. The performance of method 2 is not high even if the bins number is changed, but it might do well if features from method 2 are fused with features from method 1 for three reasons: Firstly, the texture features provide useful information about face images. Secondly, method 2 extracts only 15 features (dimension

of the feature vector is not very high). That means the fusion of the 15 features (we selected bin numbers = 5) will not add much to the overall dimensions of the feature space. Thirdly, the idea of using dynamic block size in method 2 and dynamic sub region size in method 1 can make the fusion technique perform better than with fixed sizes.

Figure 5 Accuracy of person identification in VPI database using texture features and different classifiers (method 2) with 100 block portioned image, bins number = 5, 10 (see online version for colours)

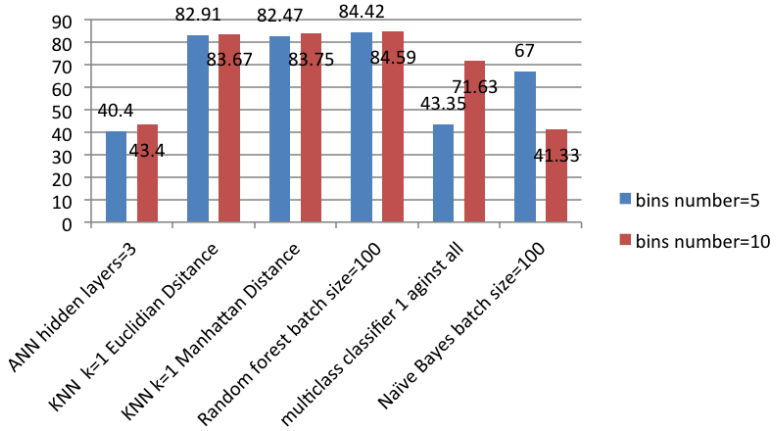
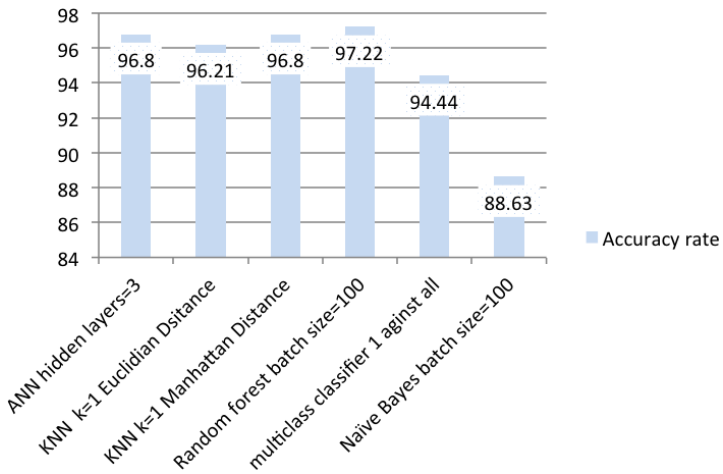


Figure 6 Accuracy rate of person identification in VPI database using fused method with five bins and 100 block portioned image (see online version for colours)

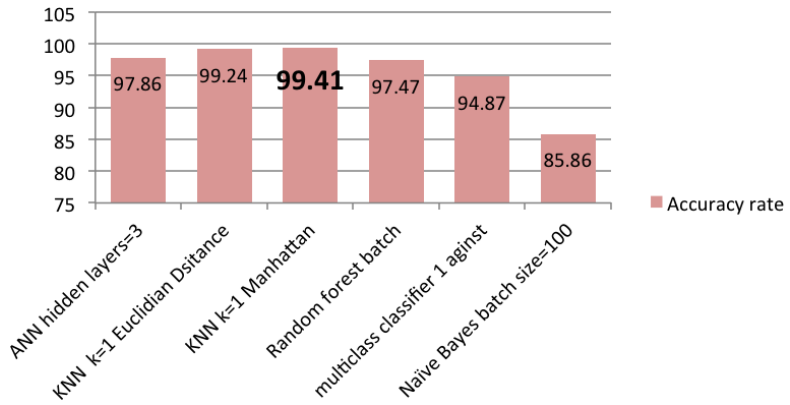


The basic concept of the fusion process in our work is the usage equal size for blocks in both geometrical and texture feature methods. For both the methods we used the same preprocessing, segmentation and image portioning processes. But, during feature extraction both of them extract features in separate and parallel way. More precisely, for each block, two pixel values are read; one from Sobel filtered image for counting edge, the other from cropped grey image to calculate the moments. Before moving to the next

block the edge ratio is valued and added to the final feature vector and the moments of this block are added to their vectors. After the image is scanned the PDF values are calculated and added to the final feature vector.

As observed in the performance and higher accuracy achieved in method 1 with respect to block numbers, we set it to 100. To benefit from texture information without exaggerating the overall feature vector size; number of bins is tuned to 5. The final feature vector of the fused method includes in total 115 features. Figure 6 illustrates the success rate of fusion of methods 1 and 2 using different classifiers. The fusion of methods 1 and 2 as seen in Figure 6 has enhanced the performance of person identification when compared to Table 2 for method 1 and Figure 5 for method 2 individually. The fusion effect appears in the highest accuracy rate; method 1 has recorded 95.96% success rate (with 100 blocks, ANN classifier – hidden layers 3), method 2 has scored 84.59% (RF classifier – ten bins). With the fusion approach, the top accuracy of 97.22% achieved with RF classifier.

Figure 7 Accuracy rate of gender classification in VPI database using fused method with five bins and 100 block portioned image (see online version for colours)



We next test the person identification accuracy for gender classification with the fused method. The fused method has achieved a highest accuracy of 99.41% of persons identification in the VPI database into male or female. The classifier KNN using k = 1 and depending on Manhattan distance has achieved this best result as demonstrated in Figure 7.

4.1 Features reduction and selection

For the purpose of dimensionality reduction, we selected the best features for person identification from the total of 115 features. We used the well-known principal component analysis (PCA) technique which provided the best 56 features. Figure 8 illustrates the results of the proposed method before and after feature selection using PCA. After performing PCA and transforming the data, the performance of the fused method has differed depending on the used classifier, however the highest accuracy has increased using ANN classifier whereas the RF classifier performance decreased.

Figure 8 Accuracy rate of Gender Classification in VPI database using fused method with five bins and 100 block portioned image (see online version for colours)

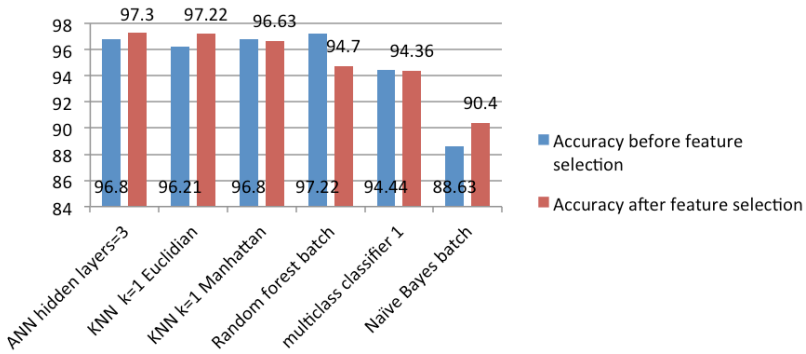


Table 3 Accuracy rate of person identification of the fused method after feature selection depending on feature ranking

Classifier #best features	*ANN hidden layers = 3*	*KNN Euclidian dist. K = 1*	*KNN Manhattan dist. K = 1*	*RF batch size = 100*	*Multiclass 1-against all*	*Naïve Bayes Batch size = 100*
110	96.8%	96.21%	96.21%	**97.55%**	94.36%	88.67%
100	96.7%	95.79%	95.96%	97.39%	94.87%	88.79%
95	96.89%	95.87%	96.04%	97.31%	93.86%	88.64%

Another feature selection experiment has been applied. We ranked the feature in order to discard the redundant features. We selected the best features and keeping the performance of the method. Table 3 describes the relationship between the number of selected features and the performance, and as can be seen, the performance is stable with features 95 to 110 for all the classifiers.

4.2 Comparing results with other approaches

We compared the proposed method with some of the well-known approaches in the field of face recognition. In what follows we describe briefly the evaluated approaches and the corresponding results using them in person identification on the VPI database.

1 *Local binary pattern:* LBP is widely used as a local texture descriptor in many image classification problems. The original LBP operator forms labels for the image pixels by thresholding the 3×3 neighbourhood with the centre value and considering the result as a binary number. The histogram of these $2^8 = 256$ different labels can then be used as an image descriptor (Li and Jain, 2011). In our experiment each pixel in the ROI was compared to each of its 8 circular neighbours to form an 8-digit binary number in which digit value = 0 if the centre pixel's value is greater than the neighbour's value, otherwise digit value = 0. as shown in Figure 9. The binary number is converted to decimal, then the histogram is computed, over the ROI, of the frequency of each 'number' occurring (i.e., each combination of which pixels are smaller and which are greater than the centre). This histogram forms a 256-dimensional final feature vector.

2 *Discrete cosine transform:* In our experiments, the DCT features are extracted from grey image. ROI is subdivided into non-overlapping blocks of 8×8 pixels. Then the DCT coefficients are computed using standard equations of DCT in two dimensions. To select the features, the first 3 zigzag values from each block are added to the feature vector.

3 *Wavelet transform (WT)*: WT is used in the application of face detection or recognition. When the image is transformed by 2D WT it is decomposed into 4 sub-bands denoted by low-low (LL); high-low (HL); low-high (LH); high-high (HH). Each sub-band can be thought of a smaller version of the image representing different properties (Li et al., 2015). In this comparison we applied one level 2D WT to the ROI of the grey image as seen in Figure 10.

We used WT for features extraction in two different methods:

a WT is combined with LBP. In this method the ROI is divided into 4 quarters, LBP feature vectors are extracted from each quarter, and then all of the four vectors are concatenated to form 1024 dimension final feature vector.

b The other method applies the described DCT method on the WT transformed image in order to form the DCT vector.

Figure 9 Standard LBP operator: left example, middle thresholding, and right the resulting pattern

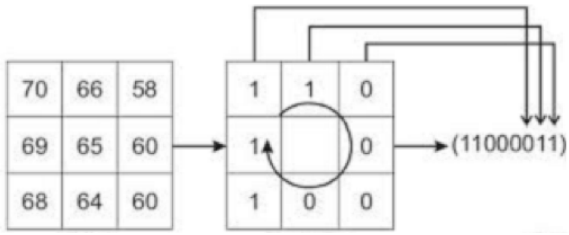


Figure 10 A one level 2D decomposition based on the wavelet transform, (a) original image from VPI (b) wavelet transform based decomposition ROI obtained after preprocessing (see online version for colours)

(a) (b)

Figure 11 The highest success rate for person identification of compared approaches (see online version for colours)

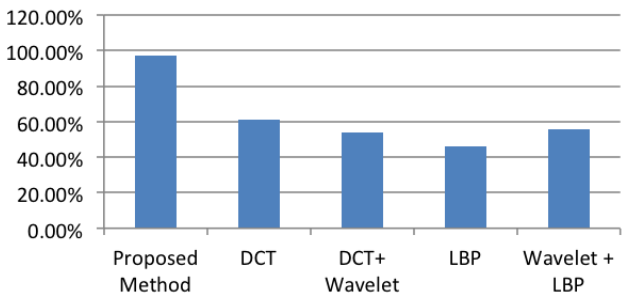


Table 4 Accuracy rate of different methods for person identification in VPI database.

Method	*# features*	*ANN*	*KNN Euclidian*	*KNN Manhattan*	*RF*	*Naïve Bayes*
Proposed	115	96.80%	96.21%	96.80%	**97.22%**	88.63%
DCT	160	57.24%	59.43%	61.36%	61.70%	42.51%
DCT + Wavelet	150	46.18%	43.96%	48.06%	53.96%	31.38%
LBP	255	14.62%	37.04%	43.69%	45.88%	26.10%
Wavelet + LBP	1,024	39.81%	43.35%	55.89%	55.44%	33.59%

Figure 11 illustrates briefly the highest accuracy of each of the compared methods. Table 4 demonstrates that the proposed method in our work has achieved the best result in terms of accuracy and feature numbers. Our method has succeeded with the highest accuracy rate (97.22%) and has gotten it with the minimal number of extracted features (115). The proposed method has carried out the best performance on VPI database in comparison with other approaches for the following reasons:

1 the compared approaches have been used in face recognition from the whole face image and as we know they have not been used to identify persons from eyes and surroundings

2 all the compared approaches are texture descriptors, on the other side, the ROI in the VPI images is small; so the texture information might not be quite enough to identify persons

3 the combination of geometrical information (edge ratio) and textural information (moments PDF) in the proposed method plays an important role in extracting the distinctive features.

5 Conclusions

Several motivations push some people to veil their faces. Most of studies in the field of face recognition are based on the whole face image being available for identification or authentication processes. In this paper, we have presented a new method to identify veiled-persons and gender classification. Our proposed method fused the local textural

features (colour moments) with the local geometrical features and we tested different classifiers. The proposed method is compared to other approaches from the literature on VPI database that is created for this study. In comparison with other approaches used to recognise faces, the proposed method is the best in terms of accuracy and with less number of features.

At the end of this study we deduced that: firstly; it is possible to identify persons from the eyes and surrounding areas. Secondly; male face is different from female face particularly in the eyes and eyebrow areas, and perhaps the thick hair, all this made it easy to distinguish between males and females. Thirdly, the proposed method does well on frontal face images, but we never tested it on non-frontal images. Creating a face database in a conservative community is very challenging, particularly for those who wear veils, that why we could not make the database larger than 100 subjects, however we aim to increase the size of the database in the future to make our results more significant and the drawn conclusions more realistic. We further will try some other advanced features extraction methods in the future as well as to vary the veiled regions among face images, where the performance of the proposed image processing methods needs to be benchmarked with other techniques.

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Notes

1 Available from the authors by request.

2 See http://www.face-rec.org/databases/ http://www.face-rec.org/databases/ for the databases and their descriptions.

3 http://www.itl.nist.gov/iad/humanid/feret/feret_master.html;

4 http://white.stanford.edu/dilaro/ee3684/code/male.zip; http://white.stanford.edu/dilaro/ee3684/code/male.zip.

5 See for example the Berkeley Segmentation DataSet (BSDS500) which consist of five human drawn contours taken as ground-truth in benchmarking segmentation methods. https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/resources.html